

MENDSPEECH REVISED LEARNING SYSTEM

Week 8: Research Capstone: Cascaded Versus Direct Repair

Freeze the benchmark, run controlled ablations, compare architectures, and publish a reproducible result.

Six focused build days plus one consolidation day. Each day ends with named artifacts or measured evidence.

Week milestone

Freeze the benchmark, run controlled ablations, compare architectures, and publish a reproducible result.

Week map

Day	Focus	Minimum evidence	Compute
Day 50	Freeze research questions and baselines	• Run a tiny dry run to ensure every result field is populated.	Local CPU for planning, Modal L4 for dry run
Day 51	Release SpeechDamageBench v1 and freeze evaluation	• Reinstall the package in a clean environment. • Regenerate a sample from the manifest and verify its checksum. • Validate that clean references remain unchanged.	Local CPU
Day 52	Run recognition and context ablations	• Plot WER versus latency and mark Pareto efficient points.	Modal L4, keep hardware fixed
Day 53	Run cascaded repair and seam ablations	• Test whether repairing more audio always helps intelligibility. • Test whether boundary matching reduces seam artifacts without materially increasing latency. • Keep recognition outputs fixed for the stitching comparison so only the repair method changes.	Modal L4
Day 54	Compare against direct latent or codec audio inpainting	• Compare raw damaged audio, MendSpeech V1 cascaded repair, full resynthesis, and the direct audio baseline on the same cases. • Select at least ten worst or most revealing cases and inspect them manually. • Identify at least one regime where each approach has an advantage, or explicitly report if the data does not support that conclusion.	Modal L4 or the smallest GPU that can run the chosen pretrained baseline, plus local analysis
Day 55	Write the research report and reproducibility guide	• Audit every major claim against a concrete table, figure, or experiment result. • Remove or soften any conclusion that is not directly supported by frozen evidence. • Verify that the report distinguishes measured facts from hypotheses and future work.	Local CPU
Day 56	Final product, demo, and clean reproduction	• Record a concise demo and create a final architecture diagram. • Reproduce one benchmark end to end from the documented command. • Verify that every public chart can be regenerated from saved result files.	Modal L4 for inference, local CPU for interface and analysis

Reference spine

- Your frozen protocol and prior results
- A reproducible pretrained direct latent or codec audio inpainting baseline
- Primary papers only when needed to interpret a result

Freeze research questions and baselines

Compute

Local CPU for planning, Modal L4 for dry run

LEARN

- Primary question: can selective semantic repair improve intelligibility while preserving more original speech than full resynthesis?
- Secondary question: can uncertainty guided context allocation improve the latency versus accuracy operating point?
- Architecture question: when does cascaded ASR plus TTS repair beat or lose to a pretrained direct latent or codec audio inpainting baseline?
- Define null outcomes, failure criteria, and claims you will not make.

BUILD IN MENDSPEECH

- Freeze code revision, model revisions, datasets, hardware, corruption configs, and metrics.
- Define baselines: raw damaged audio, full resynthesis, MendSpeech V1 cascaded repair, fixed context, adaptive context, and one pretrained direct audio inpainting baseline if reproducible.
- Do not train the direct inpainting model from scratch. The purpose is architectural comparison, not a second major training project.

EXPERIMENT AND MEASURE

- Run a tiny dry run to ensure every result field is populated.

REQUIRED OUTPUT

- experiments/capstone_protocol.md
- configs/capstone_frozen.yaml
- docs/baseline_definitions.md

Completion check: Another engineer could reproduce the protocol and understand exactly which claims compare cascaded repair with direct audio repair.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

REFERENCE PRIORITY

- Your frozen protocol and prior results
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Release SpeechDamageBench v1 and freeze evaluation

LEARN

- Severity grids.
- Speaker separated evaluation.
- Seed control and deterministic manifests.
- Package versioning and reproducibility.
- Clean regression cases that must remain untouched.

BUILD IN MENDSPEECH

- Finalize the independent SpeechDamageBench package with noise, clipping, bandwidth, dropout, and reverberation presets.
- Generate the frozen test matrix and lock manifest checksums.
- Add an installation command and a one command example that reproduces one benchmark item.

EXPERIMENT AND MEASURE

- Reinstall the package in a clean environment.
- Regenerate a sample from the manifest and verify its checksum.
- Validate that clean references remain unchanged.

REQUIRED OUTPUT

- speechdamagebench/
- speechdamagebench/README.md
- speechdamagebench/CHANGELOG.md
- benchmarks/speechdamagebench_manifest.csv
- benchmarks/README.md

Completion check: SpeechDamageBench is independently installable, deterministic, versioned, and usable without MendSpeech.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

REFERENCE PRIORITY

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Run recognition and context ablations

LEARN

- Fixed lookahead comparison.
- Adaptive context policy.
- WER, latency, RTF, memory, confidence behavior.

BUILD IN MENDESPREECH

- Run every streaming condition on the exact same benchmark subset.
- Repeat timing runs enough to estimate variance.
- Record GPU type and environment automatically through the Modal runner.

EXPERIMENT AND MEASURE

- Plot WER versus latency and mark Pareto efficient points.

REQUIRED OUTPUT

- results/capstone_streaming.csv
- results/streaming_pareto.png

Completion check: You can say whether adaptive context helped, hurt, or made no meaningful difference.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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Run cascaded repair and seam ablations

LEARN

- Repair threshold.
- Repair span padding.
- Preserve percentage.
- Full resynthesis baseline.
- Boundary energy matching, crossfade choice, and seam artifact rate.

BUILD IN MENDSPEECH

- Run Preserve, Balanced, Rescue, full resynthesis, naive selective stitching, and boundary matched selective stitching.
- Record original waveform retained, repair percentage, end to end latency, speaker similarity proxy, and seam metrics.

EXPERIMENT AND MEASURE

- Test whether repairing more audio always helps intelligibility.
- Test whether boundary matching reduces seam artifacts without materially increasing latency.
- Keep recognition outputs fixed for the stitching comparison so only the repair method changes.

REQUIRED OUTPUT

- results/capstone_cascaded_repair.csv
- results/repair_tradeoff.png
- results/seam_ablation.png

Completion check: You have a defensible result for the cascaded selective repair path and can separate recognition, reconstruction, and stitching effects.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

REFERENCE PRIORITY

- Your frozen protocol and prior results
- A reproducible pretrained direct latent or codec audio inpainting baseline
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Compare against direct latent or codec audio inpainting

Compute

Modal L4 or the smallest GPU that can run the chosen pretrained baseline, plus local analysis

LEARN

- Why text is an information bottleneck for prosody and acoustic continuity.
- Direct audio inpainting in latent or codec token spaces at a conceptual level.
- Fair baseline design when systems have different latency and compute profiles.
- Failure taxonomy across semantic correctness, speaker similarity, prosody, seam quality, and compute.

BUILD IN MENDSPEECH

- Select one reproducible pretrained direct audio inpainting or restoration baseline.
- Wrap it behind the same benchmark interface used by MendSpeech.
- Feed identical SpeechDamageBench cases and record the same metrics wherever they are meaningful.
- Create a failure casebook covering both architectures.
- Keep abstention active for MendSpeech when the inferred content is not reliable enough to reconstruct safely.

EXPERIMENT AND MEASURE

- Compare raw damaged audio, MendSpeech V1 cascaded repair, full resynthesis, and the direct audio baseline on the same cases.
- Select at least ten worst or most revealing cases and inspect them manually.
- Identify at least one regime where each approach has an advantage, or explicitly report if the data does not support that conclusion.

REQUIRED OUTPUT

- src/baselines/direct_audio_inpaint.py
- results/capstone_architecture_compare.csv
- results/capstone_failure_casebook.md
- results/architecture_tradeoff.png
- src/controller/abstain.py

Completion check: You can explain when the cascaded path is competitive, where it loses acoustic information, and whether direct audio repair earns its extra complexity on your benchmark.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

REFERENCE PRIORITY

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Write the research report and reproducibility guide

LEARN

- Abstract, motivation, hypotheses, method, baselines, metrics, results, limitations, ethics, and future work.
- Difference between observation and causal claim.
- How to report a negative or mixed architectural comparison honestly.

BUILD IN MENDSPEECH

- Write the complete report.
- Add exact reproduction commands and environment capture.
- Include the cascaded versus direct repair comparison as a dedicated section.
- Document seam limitations, prosody loss, and any conditions where the direct baseline is clearly stronger.
- Include plots with captions that state what changed and what stayed fixed.

EXPERIMENT AND MEASURE

- Audit every major claim against a concrete table, figure, or experiment result.
- Remove or soften any conclusion that is not directly supported by frozen evidence.
- Verify that the report distinguishes measured facts from hypotheses and future work.

REQUIRED OUTPUT

- REPORT.md
- REPRODUCE.md
- results/final_figures/
- docs/limitations_and_claims.md

Completion check: A technical reader can understand the contribution, the architectural tradeoff, and the limitations without opening the source code first.

Study method

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REFERENCE PRIORITY

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Final product, demo, and clean reproduction

Compute

Modal L4 for inference, local CPU for interface and analysis

LEARN

- Review the complete path from waveform and controlled corruption to streaming encoder, uncertainty, repair policy, cascaded reconstruction, direct audio baseline, and evaluation.

BUILD IN MENDSPEECH

- Build the final demo with upload or microphone input, controlled damage, live transcript, uncertainty heatmap, preserved versus repaired timeline, before and after playback, metrics, and architecture selection for benchmark playback.
- Show which milliseconds were preserved, reconstructed by the cascaded path, or repaired by the direct baseline.
- Reproduce one frozen benchmark from a fresh environment and tag a stable release.

EXPERIMENT AND MEASURE

- Record a concise demo and create a final architecture diagram.
- Reproduce one benchmark end to end from the documented command.
- Verify that every public chart can be regenerated from saved result files.

REQUIRED OUTPUT

- app/mendspeech_final.py
- README.md
- demos/final_demo.mp4
- docs/architecture.png
- release_notes.md
- results/reproduction_check.txt

Completion check: A new user can understand, run, and evaluate MendSpeech, SpeechDamageBench, the cascaded baseline, and the direct audio comparison, and you can defend every major design decision.

Study method

25 minutes focused reading. 65 minutes implementation or controlled experiment. 20 minutes research notebook. 10 minutes commit and explain the result aloud. When debugging is incomplete, continue the same task in the next session instead of pretending the day is finished.

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